Q1. Differentiate between the Internet and the World Wide Web. Compare HTTP's persistent vs. non-persistent connections. (2+2)

Q2. Explain the components of a URL using an example. (4)

Q3. What is the advantage that HTTPS provides over HTTP? Explain how attackers perform eavesdropping during a request-response action. (1+3)

Q4. Describe the document structure elements which are part of every webpage. (4)

Q5. What are list types supported by HTML? Create the following structure using Paragraph and list elements (1+3)

You can reach Michael at:

- Website
- Email
- Phone

Q6. Create the following table. Note that 4 columns are used in this table. (4)

	Maths	Science
Rajat	95	96
Mukesh	97	98

Q7. Explain the types of form submission methods. (4)

Q8. Illustrate the creation of dropdown menus within HTML forms, demonstrating the use of <select> and <option> tags with a practical example. (4)

Q9. Explain how to create reset/submit buttons in HTML forms and configure their basic properties. Detail the specific actions performed when a submit button is clicked. (2+2)

Q10. Explain the concept of semantic elements in HTML5. Why were they introduced, and how do they benefit search engines and developers compared to earlier versions of HTML? (2+2)

Q11. Describe the three main ways to include CSS in an HTML document (Inline, Internal, and External Style Sheets) and state the cascading order of priority when conflicting styles are applied. (3+1)

Q12. Discuss the syntax of a CSS rule, identifying its key components: selector, property, and value. Explain how multiple properties can be set for a single selector and the importance of semicolons in CSS declarations. (2+1+1)

Q13. Elaborate on the benefits of using external style sheets, particularly for multi-page websites. Explain how external style sheets contribute to a uniform look and feel across a website and how they facilitate quick and easy site-wide style changes.(2+2)

Q14. Explain the CSS Box Model in detail, including its four main components and how they contribute to an element's total width and height. Provide an example calculation for an element's total width. (3+1)

Q15. Describe the different values of the CSS position property. (4)

Q16. Describe the different values of the text-shadow property. (4)

Q17. Explain the different ways to include JavaScript in your HTML code with examples supporting each. In each example create a simple greet function printing “Hello World” on the web page when called. (4)

Q18. Write and Explain the output produced by the following JavaScript code snippets. (6)

1.
```
var intro = “Hi, my name is Sam, my age is -  ”;
const age = 20;
console.log(intro+age);
```

2.
```
for(const i = 0;i<=10;i++){
        console.log(i);
}
```

3.
```
var j = 7;
do{
        j - - ;
        console.log(j);
}while(j);
```

Q19. Complete the function signature to get the desired output, and explain what parameters and arguments are using the same function as an example. (4)

```
function fun(num){
   //Implement this
}
```

```
var num1 = 5;
var num2 = 12
var num3 = fun(num1);  //Num3 output must be 25
var num4 = fun(num2);  //Num4 output must be 144
console.log(num3);
console.log(num4);
```

Q20. Give the output and explain the following codes: (4)

1. ```
   let num = 15;
   function increment(){
           if(num>25){
                   num++;
           }else{
                   num = num*2;
           }
   }
   increment(num);
   console.log(num);
   increment(num);
   console.log(num);
   ```

2. ```
   var random = 777;
   var prob = 3;
   function lucky(luck){
           if(luck<=2){
                   var random = 666;
           }
           console.log(random);
   }
   lucky(prob);
   ```

Q21. Describe the output of the code and explain why this happens: (4)

1. ```
   const numbers = [1,2,3,10,21];
   numbers.sort();
   console.log(numbers);
   ```

2. ```
   const arr = [1,2,3]
   delete arr[2];
   console.log(arr.length);
   ```

Q22. Explain the difference between splicing and slicing using an example. Explain each and every parameter to draw comparisons. (6)

Q23. Explain and give the output of the following code snippet: (4)

```
for (let i = 1; i <= 5; i++) {
  setTimeout(function() {
    console.log(i);
  }, i * 1000);
}
```

Q24. Create a JavaScript object which contains keys - principal, interest rate, time and a function to calculate the total amount after interest. (2)

Q25. For the given object prototype call, create a prototype. Also in the prototype add a function to print all the details in a neat and meaningful manner. (4)

```
var proto1 = new(“Samved”,20,”Lecturer”,”WT”);
```

The keys are - name,age,occupation and subject.

Output expected - My name is Samved, I am 20 years old. I am a Lecturer and I teach the subject of WT.

Q26. Describe and explain the output of the following code snippet: (4)

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width,
initial-scale=1.0">
   <title>Document</title>
</head>
<body>
   <h1 id="demo">Hi this is a demo h1</h1>
   <h1 class="demo">Hi this is another demo h2</h1>
</body>
<script>
   let head = document.querySelector("#demo");
   head.innerHTML = "<h1><i>Changed using JS</i></h1>";
</script>
</html>
```

Q27. Explain the given code snippet with its output: (4)

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h1 id="demo">Hi this is a demo h1</h1>
    <h2 class="demo" name="h1">Hi this is another demo h2</h2>
    <h3 class="demo" name="h3">Hi this is another demo h3</h3>
</body>
<script>
    let num = document.getElementsByName("h1")
    console.log(num.length)
</script>
</html>
```

Which tag is considered for the output?

Q28. What are HTML Events? Give a few examples and explain their functioning without any code. (4)

Q29. Explain the follow events with a code example to support each: (4)

1. click
2. keydown
3. reset
4. Mousemove

Q30. Explain event bubbling and event capturing with an example. No code is required, drawing the flow is a must. (4)

Q31. Explain methods preventDefault() and stopPropagation() with suitable examples. No code required. (4)